



Module Title

Cybersecurity and Networking

Internship Assignment Submission

IPv4 Addressing & IP Configuration

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1.Theory

1.1Question and Answers

1.What is a network address?

A network address identifies the network itself. It is not normally assigned to a host.

2. What is a broadcast address?

A broadcast address is used to send data to all hosts in the same network.

3. What is a usable host address?

A usable host address is an IP address that can be assigned to a device on a network.

4. What is the first usable IP address?

It is the IP address immediately after the network address. For example, in 192.168.1.0/24, it is 192.168.1.1

5. What is the last usable IP address?

It is the IP address immediately before broadcast address. For 192.168.10.0/24, it is 192.168.10.154

6. What does /24 mean?

/24 means the first 24 bits of IPv4 address represent the network portion.

7. What subnet mask corresponds to /24?

The subnet mask of /24 is 255.255.255.0

8. What is a default gateway?

A default gateway allows a device to communicate with destination outside its local network.

9. Why does a router interface need an IP address?

A router interface needs an IP address to communicate with devices and route traffic between IP interface.

10. What is a static IP configuration?

A static IP configuration is an IP address and other network settings that are manually assigned and normally remain predictable.

11. What does "show ip interface brief" display?

It shows a quick summary of interfaces, including their IP address, status and protocol.

12. What does "no shutdown" do conceptually on a Cisco interface?

It enables the interface so it can become operational.

13. What does interface status "up/up" generally indicate?

It generally means the interface and its line protocol are both working.

14. Why should network configuration always be verified?

Configuration should be verified to make sure it worked correctly and to find problems early.

15. Why is IPv4 configuration important in cybersecurity?

Correct IPv4 configuration is important because security tools and systems such as firewalls, IDS/IPS, SIEM, Nmap and Wireshark depend on correct IP addressing.

1.2 Theory Exercise

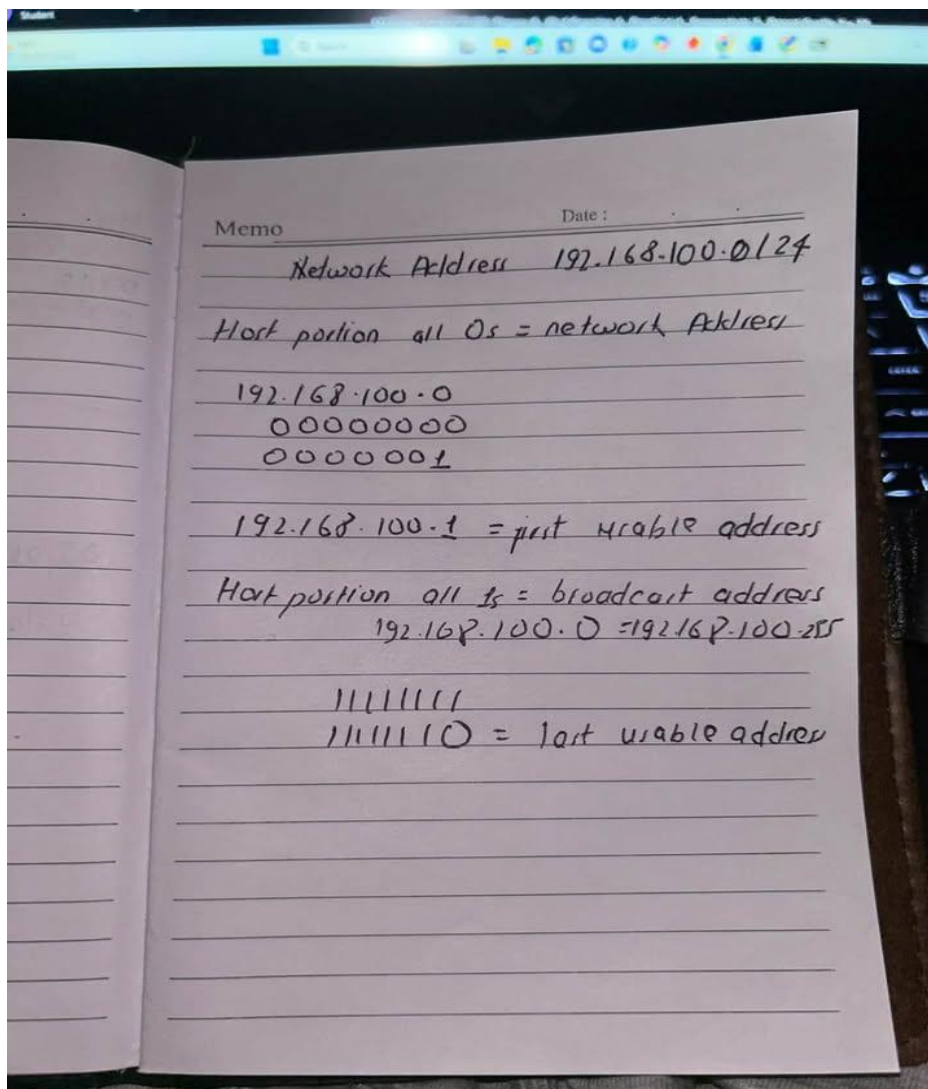
1.2.1 IPv4 for 192.168.100.0/24 identify: - Network Address - First Usable IP - Last Usable IP - Broadcast Address

Network Address: 192.168.100.0

First Usable Address: 192.168.1.1

Last Usable Address: 192.168.1.254

Broadcast Address: 192.168.1.255



2. Practical Exercise

2.1 From YOUR OWN PC (ipconfig /all) record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server.

IPv4 Address: 192.168.1.118(Preferred)

Subnet Mask: 255.255.255.0

Default Gateway :192.168.1.254

DHCP Enabled: Yes

Primary DNS: 100.127.255.165

Secondary DNS: 100.127.255.73

2.2 SCREENSHOTS Attach screenshots of: ipconfig /all · ping 127.0.0.1 · ping <your default gateway> · Packet Tracer topology (if you completed Part 4).

2.2.1 ipconfig /all → detailed adapter configuration

```
C:\Users\HP>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-7DHI07TE
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-0A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6491:ee4a:b1e9:38f5%10(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 889847847
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 1E-CE-51-F4-2D-52
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : worldlink.com.np
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 1C-CE-51-F4-2D-52
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2400:1a00:3b6d:1f0d:7575:711b:b967:f4c3(Preferred)
Temporary IPv6 Address. . . . . : 2400:1a00:3b6d:1f0d:3900:4508:5d1c:19cd(Preferred)
Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5(Preferred)
IPv4 Address. . . . . : 192.168.1.118(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, September 13, 2026 2:37:05 PM
Lease Expires . . . . . : Monday, September 14, 2026 2:37:05 PM
Default Gateway . . . . . : fe80::1%5
                          192.168.1.254
DHCP Server . . . . . : 192.168.1.254
DHCPv6 IAID . . . . . : 85773905
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
DNS Servers . . . . . : 2400:1a00:0:32::165
                          2400:1a00:8000:4::73
                          100.127.255.165
                          100.127.255.73
                          2400:1a00:0:32::165
                          2400:1a00:8000:4::73
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                          worldlink.com.np
```

2.2.2 ping 127.0.0.1 → tests the local TCP/IP loopback

```
C:\Users\HP>ping 127.0.0.1
```

```
Pinging 127.0.0.1 with 32 bytes of data:
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 127.0.0.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2.2.3 ping <your default gateway> < 192.168.1.254 >

```
C:\Users\HP>ping 192.168.1.254
```

```
Pinging 192.168.1.254 with 32 bytes of data:
```

```
Reply from 192.168.1.254: bytes=32 time=6ms TTL=64
```

```
Reply from 192.168.1.254: bytes=32 time=6ms TTL=64
```

```
Reply from 192.168.1.254: bytes=32 time=56ms TTL=64
```

```
Reply from 192.168.1.254: bytes=32 time=32ms TTL=64
```

```
Ping statistics for 192.168.1.254:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

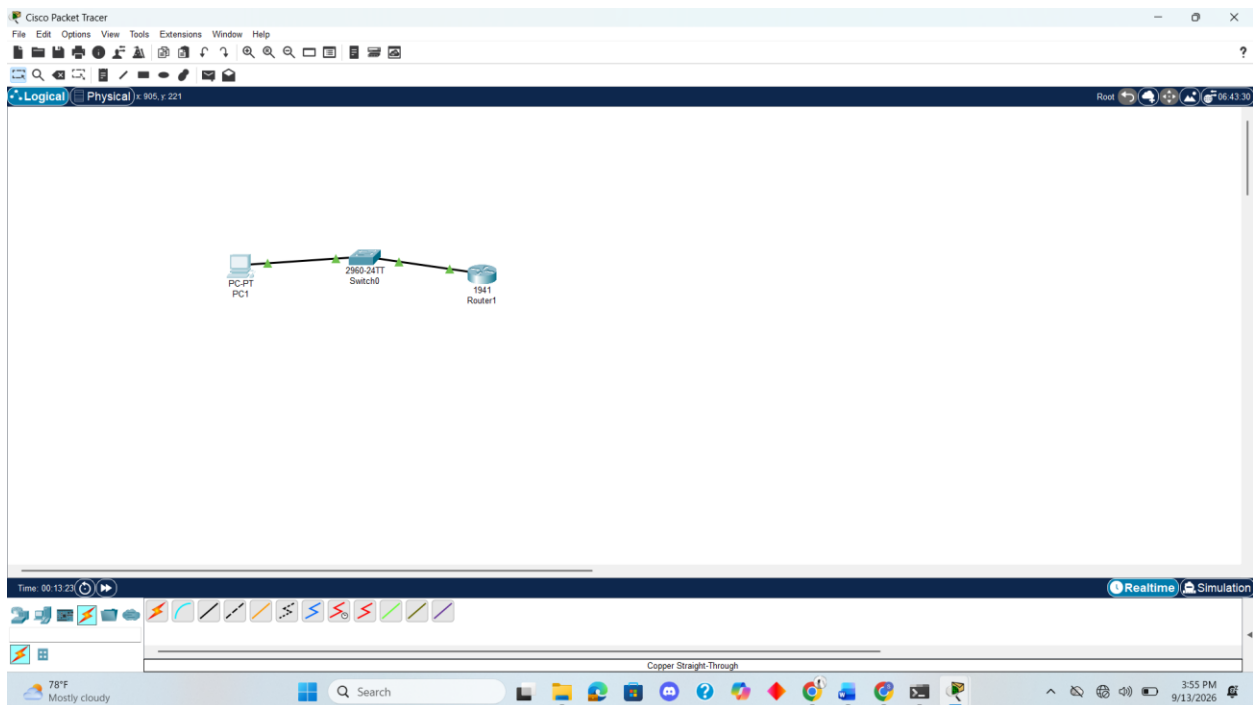
```
    Minimum = 6ms, Maximum = 56ms, Average = 25ms
```

2.3 Cisco Packet Tracer PRACTICAL

2.3.1 From YOUR OWN PC (ipconfig /all) record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server.

Topology: PC1 → Switch → Router. PC1: IPv4 192.168.10.10 · mask 255.255.255.0 · gateway 192.168.10.1. Router G0/0: 192.168.10.1/24. enable configure terminal interface gigabit Ethernet 0/0 ip address 192.168.10.1 255.255.255.0 no shutdown exits show ip interface brief Then PC1 should successfully ping 192.168.10.1. Keep this lab inside Packet Tracer or an authorized environment.

2.3.1.1 Topology



2.3.1.2 Router command

 Router1

PhysicalConfigCLIAttributes

IOS Command Line Interface

Press RETURN to get started!

Router>en
Router>enable
Router#conf t
Router#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip add
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutd
Router(config-if)#no shutdown \

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#show ip
Router(config)#show ip in
Router(config)#show ip int
Router(config)#show ip interface b
Router(config)#show ip interface br
Router(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

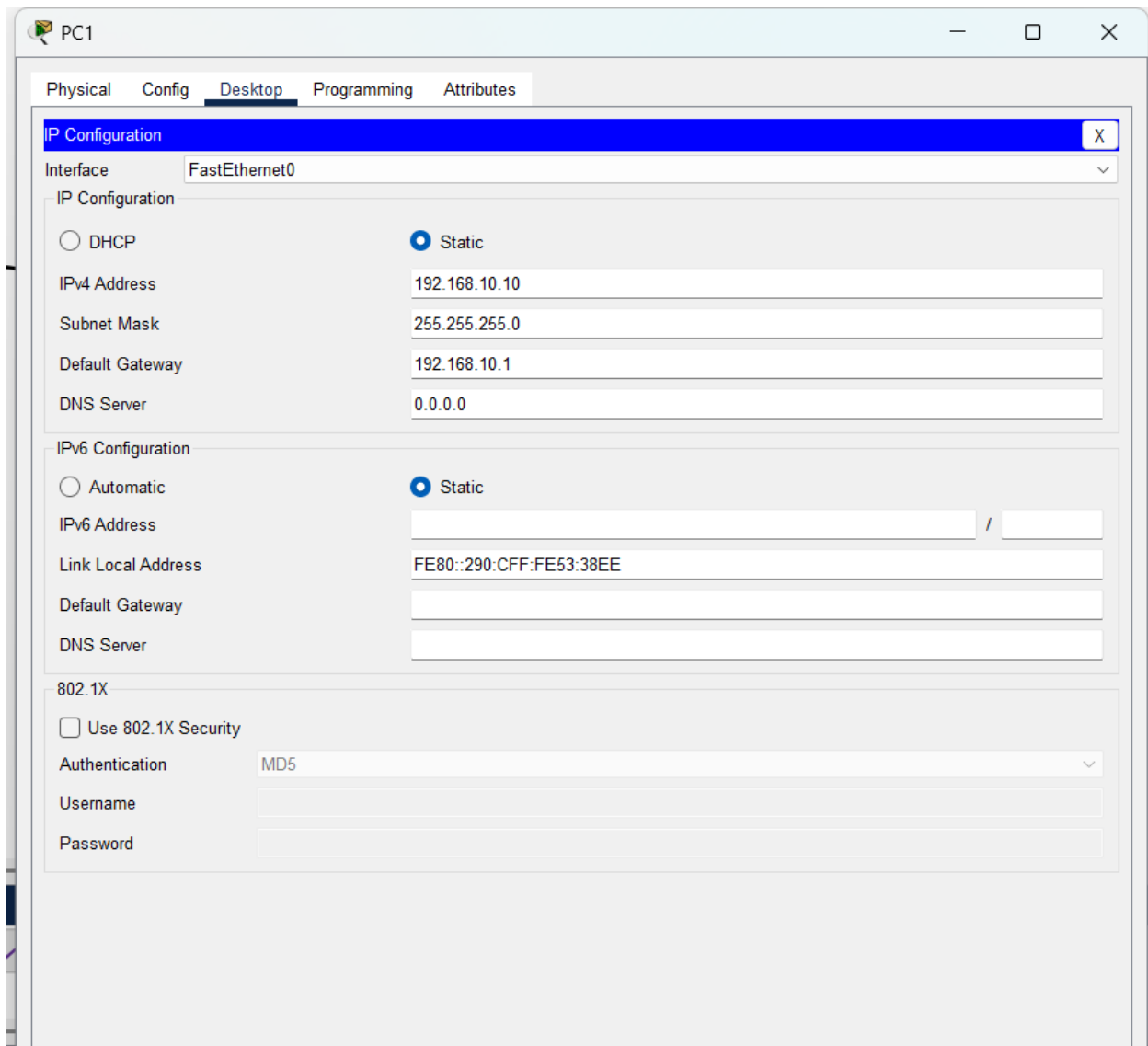
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip interface brief
InterfaceIP-AddressOK? Method StatusProtocol
GigabitEthernet0/0192.168.10.1YES manual upup
GigabitEthernet0/1unassignedYES unset administratively down down
Vlan1unassignedYES unset administratively down down
Router#

CopyPaste

☐ Top

2.3.1.3 PC1 IP Configuration



The screenshot shows a window titled "PC1" with a tabbed interface. The "Desktop" tab is selected, displaying the "IP Configuration" window. The "Interface" dropdown is set to "FastEthernet0". Under "IP Configuration", the "Static" radio button is selected. The fields are filled with: IPv4 Address: 192.168.10.10, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.1, and DNS Server: 0.0.0.0. Under "IPv6 Configuration", the "Static" radio button is also selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::290:CFF:FE53:38EE, Default Gateway (empty), and DNS Server (empty). Under "802.1X", the "Use 802.1X Security" checkbox is unchecked, and the "Authentication" dropdown is set to "MD5". The "Username" and "Password" fields are empty.

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::290:CFF:FE53:38EE

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

2.3.1.4 Test connection of PC1 with Router

